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Problem Set Number: #3

Due Date: 02/24/2026

Acknowledgement: I consult the lecture note and textbook.

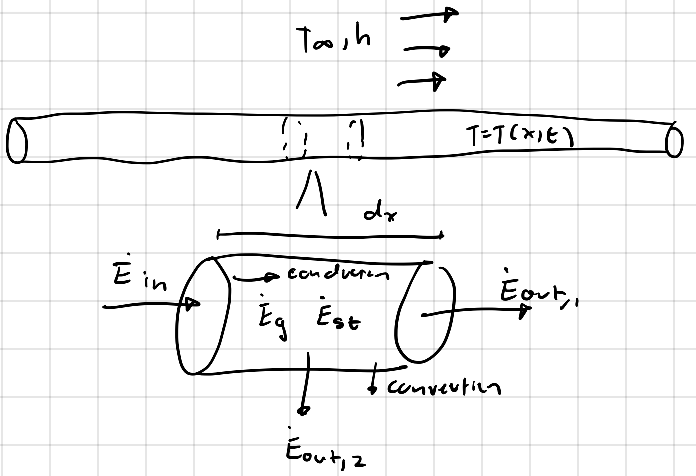
1. Known: Long-Rod : Radius $= R$, $A = \pi R^2$, $P = 2\pi R$

Find: a) Energy balance equation on a control volume in term of T

b) Governing equation for $T(x,t)$

c) Boundary equation for : $T(0,t) = T_1$, end at $x = L$, $T(x,0) = T_i$

Schematic:



Assumptions: → 1D conduction along x only

→ Convection at side surfaces

→ No radiation

→ Constant k, ρ, c_p

Properties: $\alpha = \dfrac{k}{\rho c_p}$

Analysis: a) $\boxed{\dot{E}_{in} - \dot{E}_{out} + \dot{E}_g - \dot{E}_{conv} = \dot{E}_{st}}$

Conduction: $\boxed{\dot{E}_{in} = -kA \dfrac{\partial T}{\partial x}\Big|_x \qquad \dot{E}_{out} = -kA \dfrac{\partial T}{\partial x}\Big|_{x+dx}}$

Volumetric Generation: $\boxed{\dot{E}_g = \dot{q} A\, dx}$

Convection loss from sides: $\boxed{\dot{E}_{conv} = h(2\pi R)\, dx\, (T(x,t) - T_\infty)}$

Storage: $\boxed{\dot{E}_{st} = \rho c_p (A\, dx) \dfrac{\partial T}{\partial t}}$

b) $\underbrace{(-kAT_x)_x - (-kAT_x)_{x+dx}}_{-\frac{\partial}{\partial x}(-kAT_x)dx} + \dot{q}A\, dx - hP\, dx\, (T - T_\infty) = \rho c_p A\, dx\, T_t$

↓

$kA \dfrac{\partial^2 T}{\partial x^2} dx + \dot{q}A\, dx - hP\, dx\, (T - T_\infty) = \rho c_p A\, dx\, T_t$

$$\boxed{\rho c_p \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2} + \dot{q} - \frac{hP}{A}(T - T_\infty)}$$

c) At $x=0$, constant Temperature: $T(0,t)=T_1$

At $x=L$: $-k\frac{\partial T}{\partial x}\Big|_{x=L} = h\left(T(L,t)-T_\infty\right)$

Initial condition : $T(x,0)=T_i$ for $0 \le x \le L$

2. Known: Wall A: thickness $L_A = 30cm = 0.3m$, left surface insulated

Wall B: thickness $L_B = 10cm = 0.1m$, $k_B = 30\ W/mK$, no generation

$T_\infty = 20°c$, $h = 500\ W/m^2K$, $T_w = 115°c$, $T_i = 100°c$

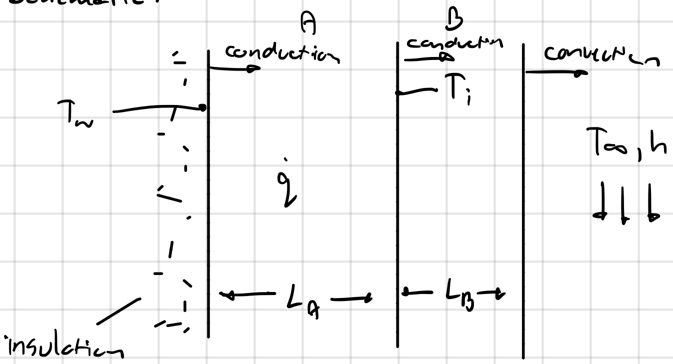
Find: a) Right surface temperature of wall B: T_s

b) $\dot{q}$ for A

c) k_A

d) Temperature profile across A and B

Schematic:



Assumptions: → Steady State

→ 1-D conduction

→ Constant properties

→ Radiation Neglected

Properties: $k_B = 30W/mK$, $k_A = ?$

$h = 500\ W/m^2K$, Wall A $\dot{q} = ?$

$T_\infty = 20°c$

Analysis:

a) Conduction through B: $q'' = \dfrac{k_B(T_i - T_s)}{L_B}$

Convection at surface: $q'' = h(T_s - T_\infty)$

equal

$$\frac{k_B T_i}{L_B} - \frac{k_B T_s}{L_B} = hT_s - hT_\infty$$

$$T_s\left(\frac{k_B}{L_B} + h\right) = \frac{k_B T_i}{L_B} + hT_\infty$$

$$T_s = \frac{\frac{k_B T_i}{L_B} + hT_\infty}{\frac{k_B}{L_B} + h} = \frac{\frac{30 \cdot 100}{0.1} + 500(20)}{\frac{30}{0.1} + 500} = \boxed{50°c}$$

b) Conduction through B:

$$q'' = \frac{k_B(T_i - T_s)}{L_B} = \frac{30(100-50)}{0.1} = 15000\ W/m^2$$

$$q'' = \dot{q} L_A$$

$$\dot{q} = \frac{q''}{L_A} = \frac{15000}{0.3} = \boxed{50000\ W/m^3}$$

c) $\frac{\partial^2 T}{\partial x^2} = \frac{-\dot{q}}{k_A}$

$\frac{dT}{dx} = \frac{-\dot{q}}{k_A} x + C_1$

IC: $x=0,\ \frac{\partial T}{\partial x} = 0$

$C_1 = 0$

$T = \frac{-\dot{q}}{2k_A} x^2 + C_2 \rightarrow C_2 = T_w$

$T(0) = T_w$

$T_A(x) = T_w - \frac{\dot{q}}{2k_A} x^2$

Plug in $T(L_A) = T_i$:

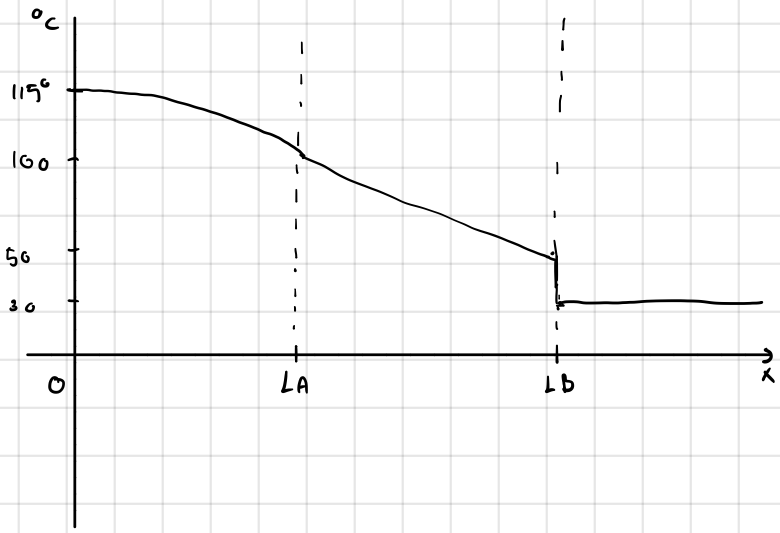
$T_i = T_w - \frac{\dot{q}}{2k_A} (L_A)^2$

$T_w - T_i = \frac{\dot{q}}{2k_A} (L_A)^2$

$\therefore k_A = \frac{\dot{q} L_A^2}{2(T_w - T_i)} = \frac{(50000)(0.3)^2}{2(115-100)} = \boxed{150\ W/mK}$

d) For $0 \leq x \leq L_A$: $T_A(x) = T_w - \frac{\dot{q}}{2k_A} x^2$

At 0 $T(0) = T_w = 115°C$, At $x = LA \rightarrow T(LA) = T_i = 100°C$



From $L_A \leq x \leq L_A + L_B$

$T_B(x) = T_i - \frac{q''}{k_B} (x - L_A)$

$x = L_A,\ T = 100 \rightarrow x = L_A + L_B,\ T = 50°C$

3. Known: Hot Gas : $T_{\infty,o} = 1616K$ $h_o = 1000\ W/m^2K$

Cooling Air : $T_{\infty,i} = 400K$ $h_i = 500W/m^2K$

Superalloy blade : $k_b = 25\ W/mK$ $L_b = 5mm = 0.005m$, $T_{max} = 1700K$

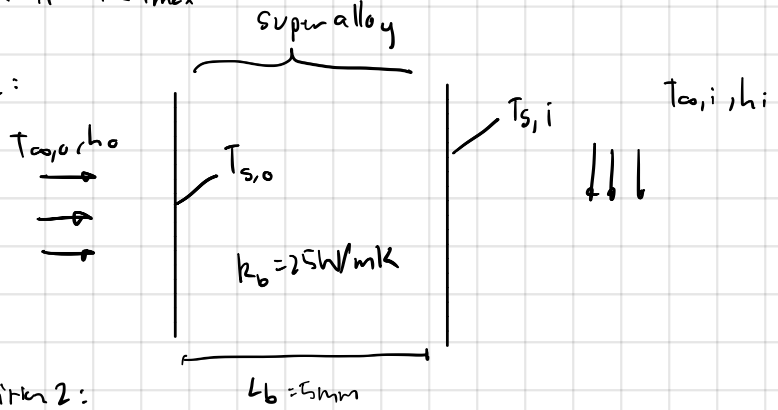
TBC ceramic : $k_{TBC} = 1W/mK$ $L_{TBC} = 0.5\ mm = 0.0005\ m$

$R'' = 10^{-4}\ m^2/KW$

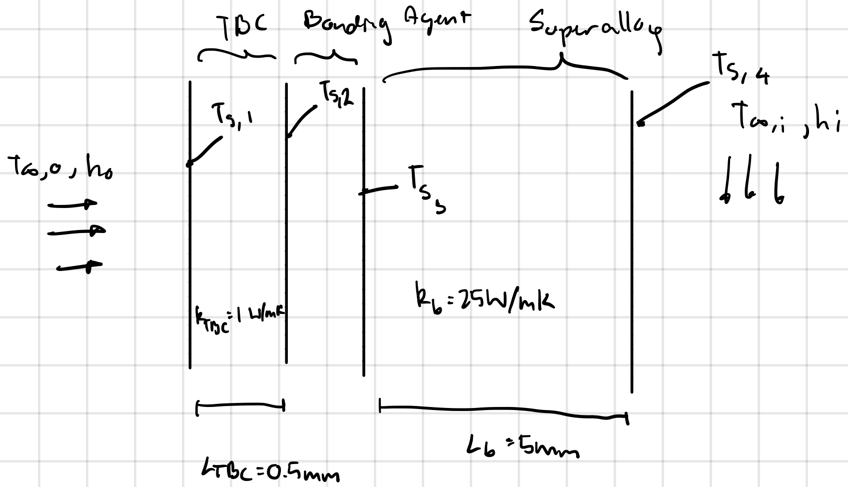
Find : a) Condition 1 (no TBC) : draw resistence network, compute exterior superalloy surface temperature and check $T \leq T_{max}$

b) Condition 2 (with TBC) : draw thermal resistent network, calculate temperature of alloy blade and check if $T \leq T_{max}$

Schematic:



Condition 2:



Assumptions: → Steady State

→ 1D heat transfer

→ Constant k, h

→ Radiation Neglected

Properties: Convection (Hot) = $R''_{h_o} = \frac{1}{h_o} = \frac{1}{1000} = 0.001\ m^2K/W$

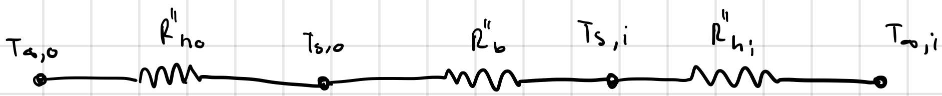
Convection (Cold) = $R_{h_i} = \frac{1}{500} = 0.002\ m^2K/W$

Superalloy conduction: $R''_b = \frac{L_b}{k_b} = \frac{0.005}{25} = 0.0002\ m^2K/W$

TBC conduction: $R''_{TBC} = \frac{L_{TBC}}{k_{TBC}} = \frac{0.0005}{1} = 0.0005\ m^2K/W$

Bonding Agent: $R'' = 0.0001 m^2K/W$

Analysis: a) No TBC:



Total Resistance: $R''_{total} = R''_{ho} + R''_{b} + R''_{hi}$

$= 0.001 + 0.0002 + 0.002 = 0.0032 \, m^2K/W$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{tot,1}} = \frac{1616 - 400}{0.0032} = \frac{1216}{0.0032} = 380\,000 \, W/m^2$$

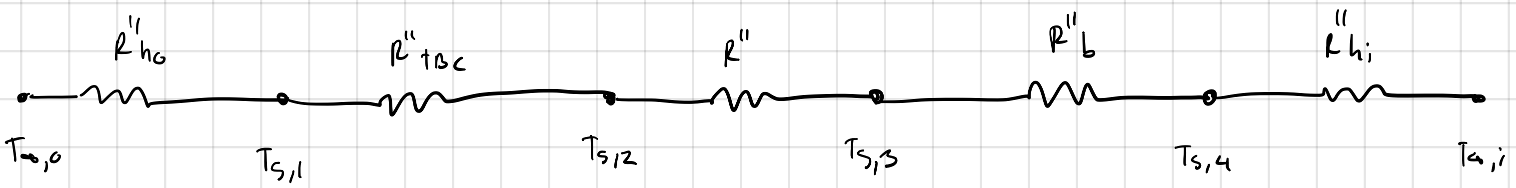
Find $T_{s,o}$: $T_{\infty,o} - T_{s,o} = q'' R''_o = q''\left(\frac{1}{h_o}\right)$

$T_{s,o} = T_{\infty,o} - q''\left(\frac{1}{h_o}\right)$

$= 1616 - 340000(0.001) = \boxed{1236K}$

$\therefore$ It is not safe as it exceeds $T_{max} = 1200K$

b)



$R''_{total} = R''_{ho} + R''_{TBC} + R'' + R''_{b} + R''_{hi}$

$= 0.001 + 0.0005 + 0.0001 + 0.0002 + 0.002 = 0.0038 \, m^2K/W$

$$q'' = \frac{T_{\infty,o} - T_{\infty,i}}{R''_{tot,2}} = \frac{1616 - 400}{0.0038} = \frac{1216}{0.0038} = 320000 \, W/m^2$$

Find $T_{s,3}$: $\Delta T_{ho} = q'' R''_{ho} = 320000(0.001) = 320K$

$T_{s,1} = T_{\infty,o} - \Delta T_{ho} = 1616 - 320 = 1296K$

$\Delta T_{TBC} = q'' R''_{TBC} = 320000(0.0005) = 160K$

$T_{s,2} = T_{s,1} - 160 = 1136K$

$\Delta T_{int} = q'' R'' = 320000(0.0001) = 32K$

$\therefore T_{s3} = T_{s,2} - \Delta T_{int} = 1136 - 32 = \boxed{1104K}$

$\therefore$ It is safe as it is less than $T_{max} = 1200K$